Password Protected Door Lock System Using Arduino

1. Introduction

Security is a major concern in homes, offices, and industries. Traditional lock systems can be easily duplicated or broken. To overcome this problem, an electronic password-based door lock system provides better security and flexibility.

This project implements a password-protected door locking system using Arduino, where a user must enter the correct password using a keypad to unlock the door. The door is controlled by a servo motor, and system status is displayed on a 16×2 LCD.

2. Objective of the Project

To design a secure door lock system using a keypad-based password.

To control door locking and unlocking using a servo motor.

To display system messages such as password status and door state on an LCD.

To provide an automatic closing mechanism after a fixed time.

3. Components Used

1. Arduino Uno - Main microcontroller
2. 4×4 Keypad - Used for password input
3. 16×2 LCD Display - Displays system messages
4. Servo Motor - Controls door opening and closing
5. Potentiometer - Adjusts LCD contrast
6. Connecting Wires - Circuit connections
7. Power Supply - USB / External power

4. Block Diagram Description

- The keypad sends password input to the Arduino Uno.
- The Arduino processes the input and compares it with the predefined password.
- If the password is correct, Arduino activates the servo motor to open the door.
- If the password is wrong, access is denied.
- The LCD display continuously shows the system status such as:
- “Enter Password”
- “Door is Open” or “Wrong Password”

5. Working Principle

1. When the system is powered ON, the LCD displays “Protected Door”.

2. The user enters a 4-digit password using the keypad.

3. Each key pressed is displayed on the LCD.

4. The entered password is compared with the stored password (1234).

5. If the password is correct:

Servo motor rotates and opens the door.

LCD displays “Door is Open”.

After a fixed delay, the door automatically closes.

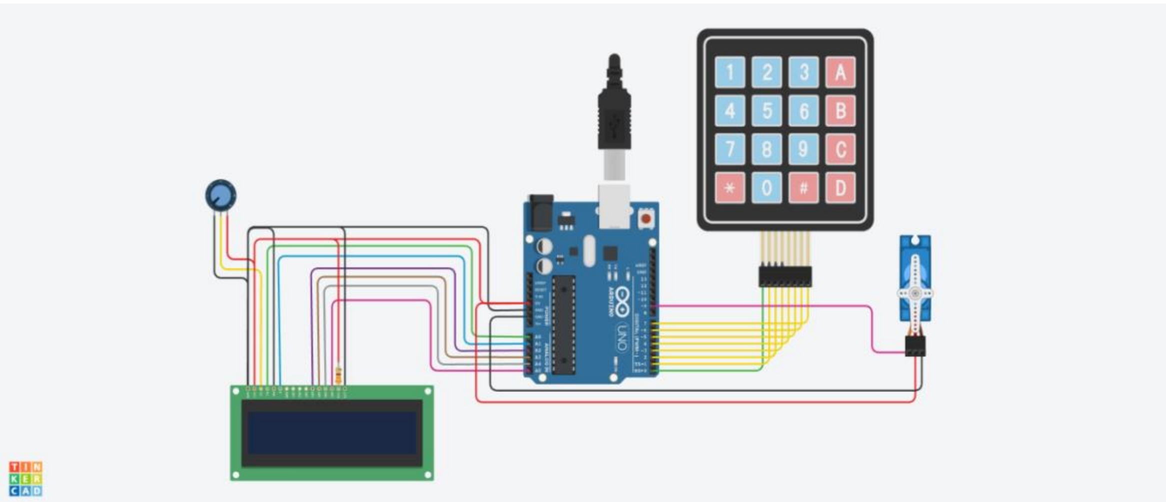
6. If the password is incorrect:

LCD displays “Wrong Password”.

Door remains locked.

7. Pressing # closes the door manually when it is open.

6. Circuit Description



7. Code:

```
#include <Keypad.h>

#include <LiquidCrystal.h>

#include <Servo.h>

#define Password_Length 5

Servo myservo;

LiquidCrystal lcd(A0, A1, A2, A3, A4, A5);
```

```
int pos = 0;
char Data[Password_Length];
char Master[Password_Length] = "1234";
byte data_count = 0;
byte master_count = 0;
bool Pass_is_good;
bool door = false;
char customKey;
/* --- preparing keypad --- */
const byte ROWS = 4;
const byte COLS = 4;
char keys[ROWS][COLS] = {
 {'1','2','3','A'},
 {'4','5','6','B'},
 {'7','8','9','C'},
 {'*','0','#','D'}
};
byte rowPins[ROWS] = {0, 1, 2, 3};
byte colPins[COLS] = {4, 5, 6, 7};
Keypad customKeypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);

/* ---------------- Main Action ---------------- */

void setup() {
 myservo.attach(9, 2000, 2400);
 ServoClose();
 lcd.begin(16, 2);
 lcd.print("Protected Door");
 loading("Loading");
 lcd.clear();
}
```

```
void loop() {
 if (door == true) {
  customKey = customKeypad.getKey();
  if (customKey == '#') {
   lcd.clear();
   ServoClose();
   lcd.print("Door is closed");
   delay(3000);
   door = false;
  }
 }
 else {
  Open();
 }
}
void loading(char msg[]) {
 lcd.setCursor(0, 1);
 lcd.print(msg);
 for (int i = 0; i < 9; i++) {
  delay(1000);
  lcd.print(".");
 }
}
void clearData() {
 while (data_count != 0) {
  Data[data_count - 1] = 0;
  data_count--;
 }
}
void ServoClose() {
 for (pos = 180; pos >= 0; pos -= 10) {
```

```
    myservo.write(pos);
    delay(15);
  }
}
void ServoOpen() {
  for (pos = 0; pos <= 180; pos += 10) {
    myservo.write(pos);
    delay(15);
  }
}
void Open() {
  lcd.setCursor(0, 0);
  lcd.print("Enter Password");
  customKey = customKeypad.getKey();
  if (customKey) {
    Data[data_count] = customKey;
    lcd.setCursor(data_count, 1);
    lcd.print(Data[data_count]);
    data_count++;
  }
  if (data_count == Password_Length - 1) {
    if (!strcmp(Data, Master)) {
      lcd.clear();
      ServoOpen();
      lcd.print("Door is Open");
      door = true;
      delay(5000);
      loading("Waiting");
      lcd.clear();
      lcd.print("Time is up!");
      delay(1000);
```

```
   ServoClose();
   door = false;
  }
  else {
   lcd.clear();
   lcd.print("Wrong Password");
   door = false;
  }
  delay(1000);
  lcd.clear();
  clearData();
 }
}
```

8. Algorithm

1. Start the system

2. Initialize LCD, keypad, and servo motor

3. Display “Enter Password”

4. Read keypad input

5. Compare entered password with stored password

6. If password matches:

Open door using servo

Display success message

7. Else:

Display error message

8. Clear data and repeat